
BACHELOR OF SCIENCE IN COMPUTER SCIENCE — THESIS PRESENTATION

Deep Learning-Based Detection of Mangrove Gain and Loss in Barangay Dulao, Aringay, La Union Using U-Net and Sentinel-2 Imagery

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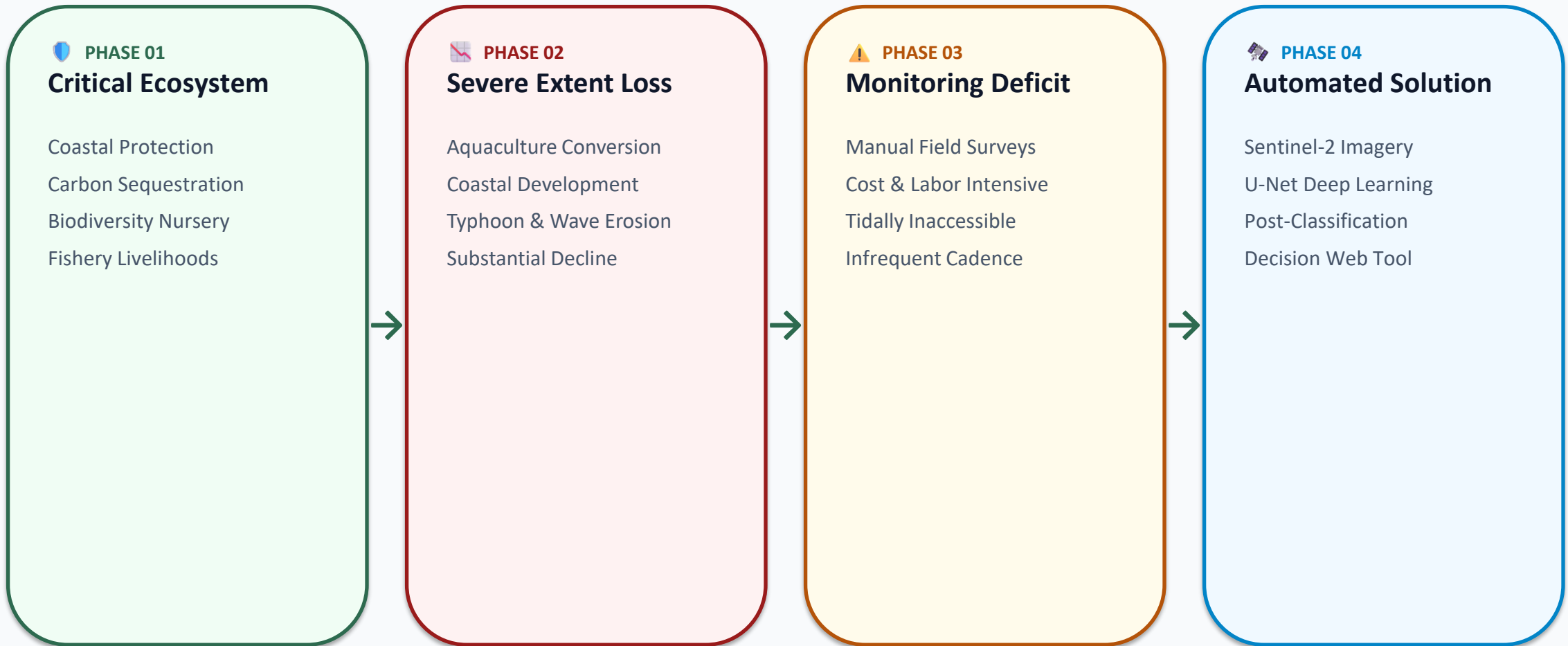
INSTITUTION & PARTNER

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Agoo, La Union, Philippines

In Collaboration with:

Municipal Environment & Natural Resources Office
(MENRO), Aringay, La Union

Ecological Importance & The Imperative for Mangrove Monitoring



Problem Statement & The Localized Research Gap

1. Spatial Complexity & Tidal Inaccessibility

Barangay Dulao's estuarine mangroves feature dense aerial root tangles and deep muddy intertidal zones. Field personnel cannot safely or exhaustively traverse the entire 316.74-hectare zone on foot to log canopy disruptions.

Result: Patchy, delayed, and incomplete coverage.

2. Failure of Traditional Pixel Classifiers

Conventional vegetation thresholding (e.g., raw NDVI thresholds alone or Maximum Likelihood) easily confuses healthy inland crops, fishpond algae, and tidal mudflats with true mangrove canopies due to overlapping spectral signatures.

Result: High false-positive rates and misclassified boundaries.

3. The Software-LGU Usability Barrier

Existing remote sensing research produces raw geospatial GeoTIFFs or Python scripts. Municipal MENRO personnel lack specialized GIS software licenses or command-line scripting skills required to extract actionable planning insights.

Result: Scientific models fail to support municipal policy.

RESEARCH GAP: Need for a systematic, localized mangrove change-detection workflow combining context-aware deep learning with an intuitive municipal decision-support web prototype.

General & Specific Research Objectives

GENERAL OBJECTIVE

To develop an automated change detection and decision-support workflow for mangrove canopy gain and loss in Barangay Dulao, Aringay, La Union from 2019 to 2024 by integrating multi-spectral Sentinel-2 satellite imagery, a 5-channel U-Net deep learning model, and an interactive web-based prototype.

SPECIFIC OBJECTIVE 1

Data Pipeline & Feature Prep

- Acquire Sentinel-2 L2A surface reflectance (2019–2024 dry season).
- Extract Green (B3), Red (B4), NIR (B8), and SWIR (B11).
- Resample B11 from 20m to 10m bilinear resolution.
- Compute and normalize NDVI to produce 5-channel 256×256 tensors.

SPECIFIC OBJECTIVE 2

U-Net Model & Evaluation

- Construct a 5-channel modified U-Net architecture.
- Train model using 80/20 train-validation partitioning.
- Optimize with Adam optimizer & Binary Cross-Entropy loss.
- Rigorously evaluate using F1-score, IoU, and mAP across a 0.05–0.95 threshold sweep.

SPECIFIC OBJECTIVE 3

PCC & Prototype System

- Execute Post-Classification Comparison (PCC) between annual pairs.
- Classify transitions into Gain, Loss, Stable, and Stable Non-mangrove.
- Develop interactive Flask/Leaflet web GIS dashboard.
- Implement automated one-click PDF reporting for Aringay MENRO.

Study Area & Temporal Setting: Barangay Dulao, Aringay

GEOGRAPHIC DELIMITATION

Barangay Dulao, Aringay, La Union

Geographic Coordinates:

120.320°E to 120.343°E

16.364°N to 16.388°N

Authoritative Area (ROI):

316.74 Hectares (25-point Polygon)

Ecological Context:

Aringay River estuary discharging into Lingayen Gulf, South China Sea

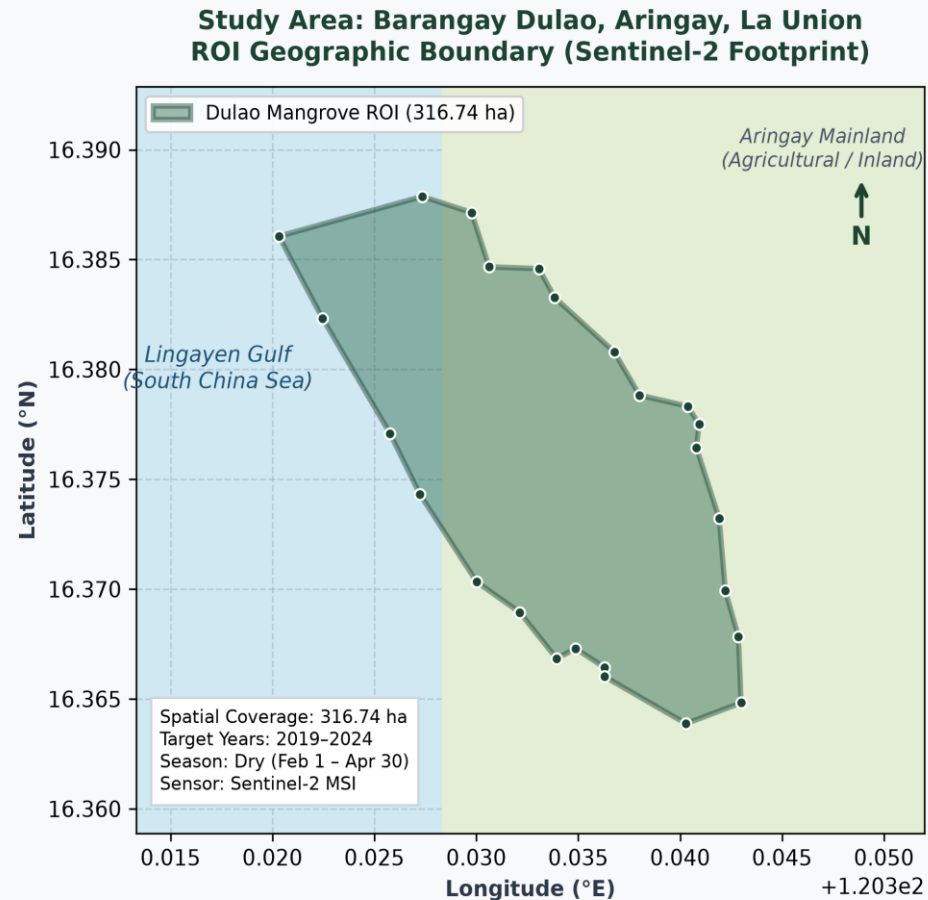
Temporal Coverage:

6 Annual Epochs: 2019, 2020, 2021, 2022, 2023, 2024

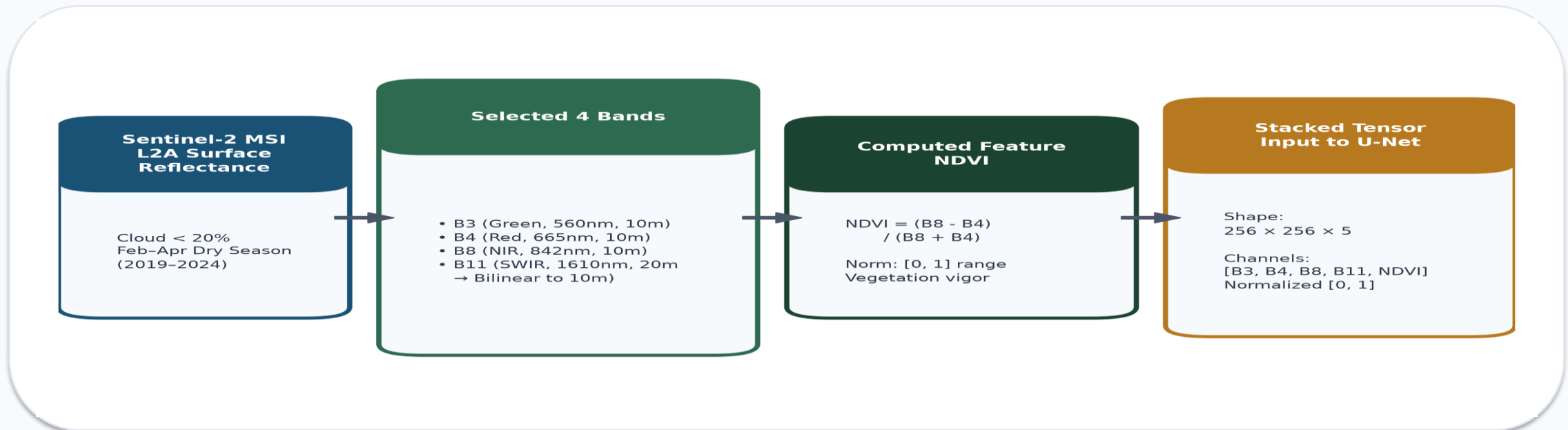
Seasonal Acquisition Window:

February 1 to April 30 (Dry Season)

Cloud cover filtered < 20% to minimize cloud shadow artifacts and tidal variations



Satellite Image Transformation: From Raw Sensor to 5D Tensor



1. Spectral Band Selection

Selects Green (B3, 560nm), Red (B4, 665nm), NIR (B8, 842nm), and SWIR (B11, 1610nm). Captures chlorophyll absorption, cellular scattering, and moisture.

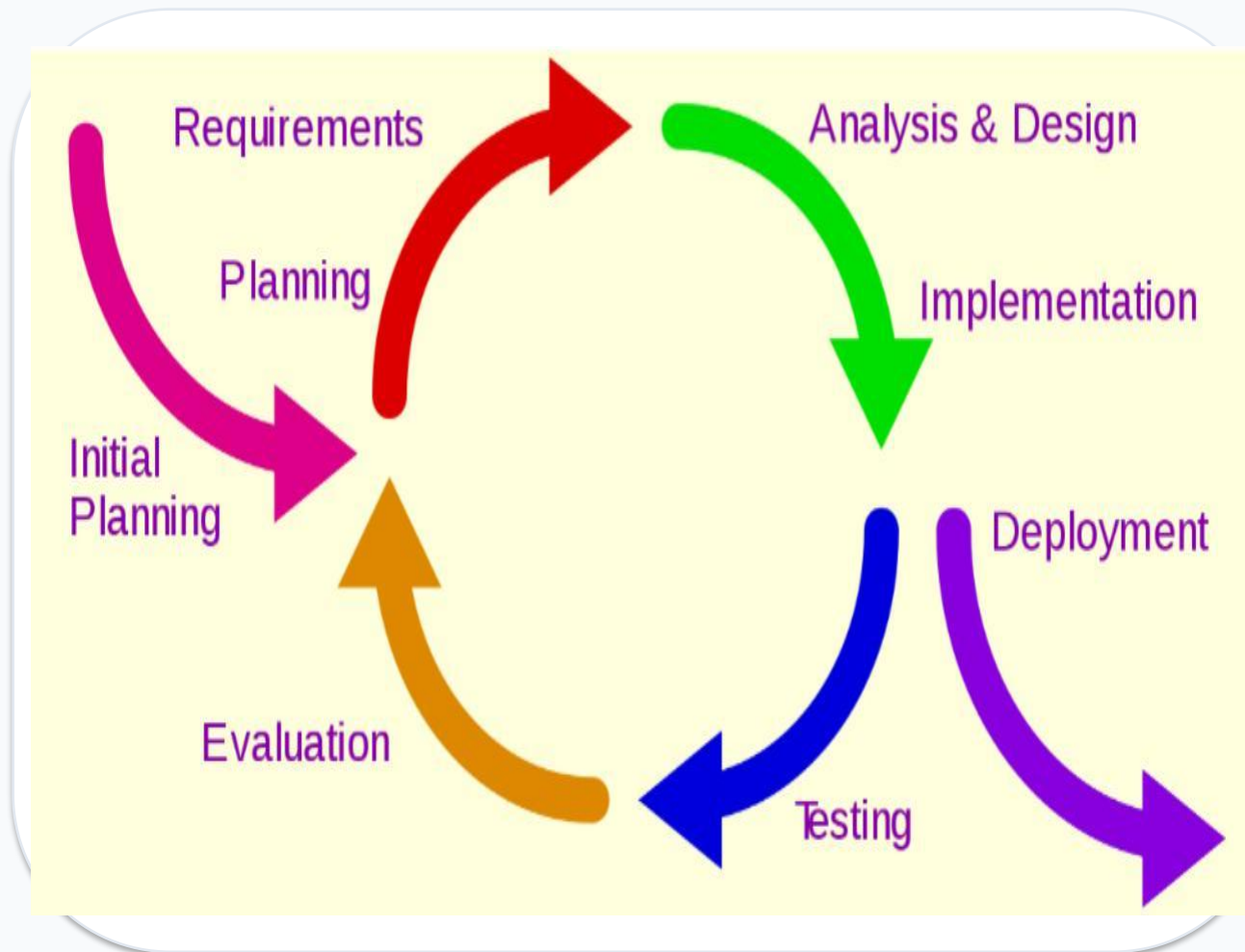
2. Spatial Harmonization

B3, B4, and B8 are native 10m pixels. B11 has native 20m resolution and is bilinearly interpolated to 10m to match the spatial grid without pixel distortion.

3. 5D Input Tensor Construction

NDVI is calculated and normalized to [0, 1]. All 5 layers are stacked into a 256×256×5 tensor with standard reflection padding for border patches.

Research Design: Iterative and Incremental Development (IID)



IID MODEL IMPLEMENTATION PHASES

- **Phase 1: Domain Analysis:** Consulted Aringay MENRO on local conservation needs.
- **Phase 2: Data Acquisition:** Extracted 2019–2024 Sentinel-2 scenes via Earth Engine.
- **Phase 3: Preprocessing & Stack:** Harmonization, bilinear resampling, and NDVI calculation.
- **Phase 4: Dataset & Partitioning:** Patch slicing (256×256) & 80/20 train-validation protocol.
- **Phase 5: U-Net Architecture:** Constructed 5-channel encoder-decoder deep network.
- **Phase 6: Training & Validation:** Adam optimizer, BCE loss, EarlyStopping (patience=10).
- **Phase 7: Change Detection (PCC):** Matrix comparison for Gain, Loss, Stable, and Other.
- **Phase 8: Web Prototype & Reports:** Interactive Flask web GIS and automated PDF reports.

Sentinel-2 MSI Spectral Band Rationale

Feature	Band Description	Central Wavelength	Native Resolution	Biophysical Role in Mangrove Detection
Band 3 (B3)	Green Reflectance	560 nm	10 meters	Vegetation reflectance peak; distinguishes green canopy from dark water bodies.
Band 4 (B4)	Red Absorption	665 nm	10 meters	Chlorophyll absorption valley; essential contrast baseline for vegetation vigor.
Band 8 (B8)	Near Infrared (NIR)	842 nm	10 meters	Maximum cellular leaf scattering; critical for delineating canopy boundary from water.
Band 11 (B11)	Short-Wave Infrared (SWIR)	1610 nm	20 m (Resampled to 10m)	Moisture and soil sensitivity; discriminates tidal mudflats and fishpond banks from canopy.
NDVI	Normalized Difference Veg.	Derived (B8 & B4)	10 meters	Structural greenness indicator; standardizes photosynthetic density across seasons.

TENSOR SPECIFICATION: 256 × 256 × 5

By stacking 4 multispectral bands [B3, B4, B8, B11] plus the computed NDVI into a unified 5-channel tensor, the U-Net receives both primary electromagnetic reflectances and higher-level biological indices simultaneously, eliminating ambiguity between tidal mudflats and mangrove cover.

Normalized Difference Vegetation Index (NDVI) Formulation

MATHEMATICAL FORMULATION

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4})$$

Normalized Range: [-1.0, +1.0] → Rescaled to [0.0, 1.0] for Neural Network Input

Chlorophyll Contrast

Capitalizes on the sharp contrast between strong chlorophyll absorption in Red (B4) and high cellular scattering in NIR (B8), isolating photosynthetic biomass.

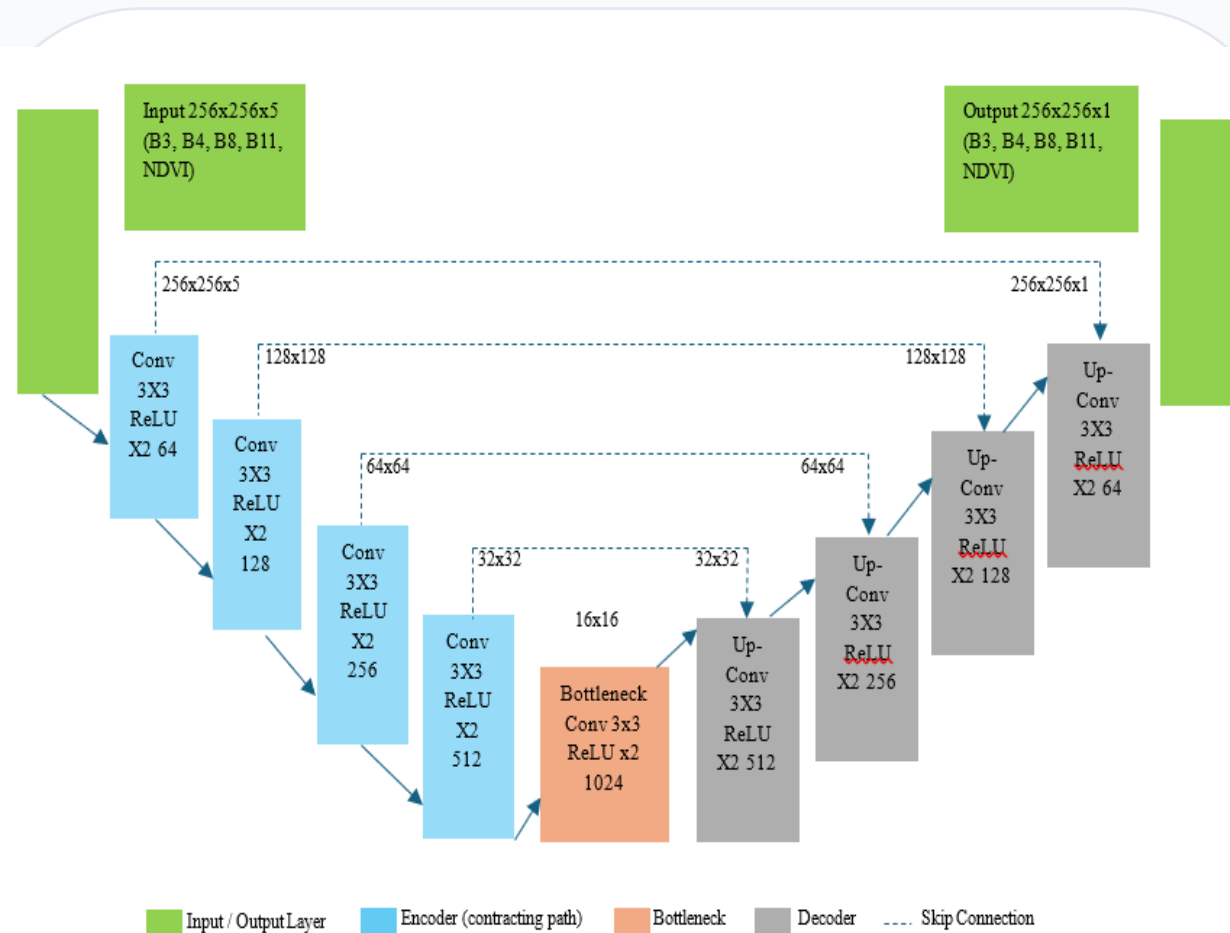
Water Body Discrimination

Open water (Lingayen Gulf, river channels) absorbs NIR strongly, yielding negative or near-zero NDVI values, cleanly separating waterways from canopy fringes.

Atmospheric Normalization

Ratio-based formulation cancels out topographic shading, solar illumination angle variations, and residual scene haze across the 2019–2024 temporal baseline.

5-Channel U-Net Architecture for Semantic Segmentation



U-NET SPECIFICATIONS (THESIS PAGE 15)

Input Dimension:

256 × 256 × 5 (B3, B4, B8, B11, NDVI)

Contracting Path (Encoder):

Successive 3×3 convolutions, ReLU activation, and 2×2 max pooling (feature extraction).

Bottleneck Layer:

Deepest representations capturing macro spatial context across Dulao's coastline.

Expansive Path (Decoder):

Up-convolutions restoring full 256×256 spatial resolution.

Skip Connections:

Directly concatenate fine-grained encoder feature maps to decoder to preserve precise mangrove boundaries.

Output Layer:

1×1 convolution with Sigmoid activation → probability map [0.0, 1.0] of mangrove presence.

Experimental Protocol: Training & Validation Methodology

DATA PARTITIONING PROTOCOL

80% Training / 20% Validation Split

Methodological Rationale:

Selected by the research team as the experimental data-partitioning protocol for model evaluation.

80% Training Set:

Used by the Adam optimizer to compute gradients and update convolutional weights via backpropagation.

20% Holdout Validation Set:

Completely isolated from model training. Evaluated at each epoch to monitor generalization and prevent overfitting.

Academic Integrity Verification:

This 80/20 partition is an established experimental choice by the research team and is not attributed to external papers.

TRAINING HYPERPARAMETERS & CONTROLS

Optimization & Regularization

Optimizer:

Adam (Adaptive Moment Estimation)

Initial Learning Rate:

0.001 (with adaptive decay)

Loss Function:

Binary Cross-Entropy (BCE) Loss

Regularization:

EarlyStopping (Patience = 10 epochs on val_loss)

Model Checkpointing:

Save only model weights achieving minimum validation loss

Threshold Evaluation Sweep:

Classification threshold swept from 0.05 to 0.95 (step 0.05) to optimize empirical F1-score

Post-Classification Comparison (PCC) Transition Matrix

Post-Classification Comparison (PCC) Transition Matrix

Pixel-by-Pixel Logic: Previous Year (T_1) → Current Year (T_2)

GAIN (Mangrove Expansion)

Previous: Non-Mangrove (0) → Current: Mangrove (1)

Natural colonization, estuarine progradation, or artificial mangrove reforestation planting.

LOSS (Mangrove Deforestation)

Previous: Mangrove (1) → Current: Non-Mangrove (0)

Fishpond conversion, typhoon/storm surge scour, infrastructure encroachment, or coastal harvesting.

STABLE MANGROVE (Persistent)

Previous: Mangrove (1) → Current: Mangrove (1)

Healthy core mangrove canopy intact across both observation years without disturbance.

STABLE NON-MANGROVE (Unchanged)

Previous: Non-Mangrove (0) → Current: Non-Mangrove (0)

Open water (Lingayen Gulf, river channels), mudflats, agricultural land, or coastal sandy substrate.

THE 4 CANOPY TRANSITION STATES

GAIN (Value = +1):

Non-Mangrove in T_1 → Mangrove in T_2 . Indicates canopy recovery, natural regeneration, or planting.

LOSS (Value = -1):

Mangrove in T_1 → Non-Mangrove in T_2 . Indicates clearing, fishpond expansion, or storm scour.

STABLE MANGROVE (Value = 2):

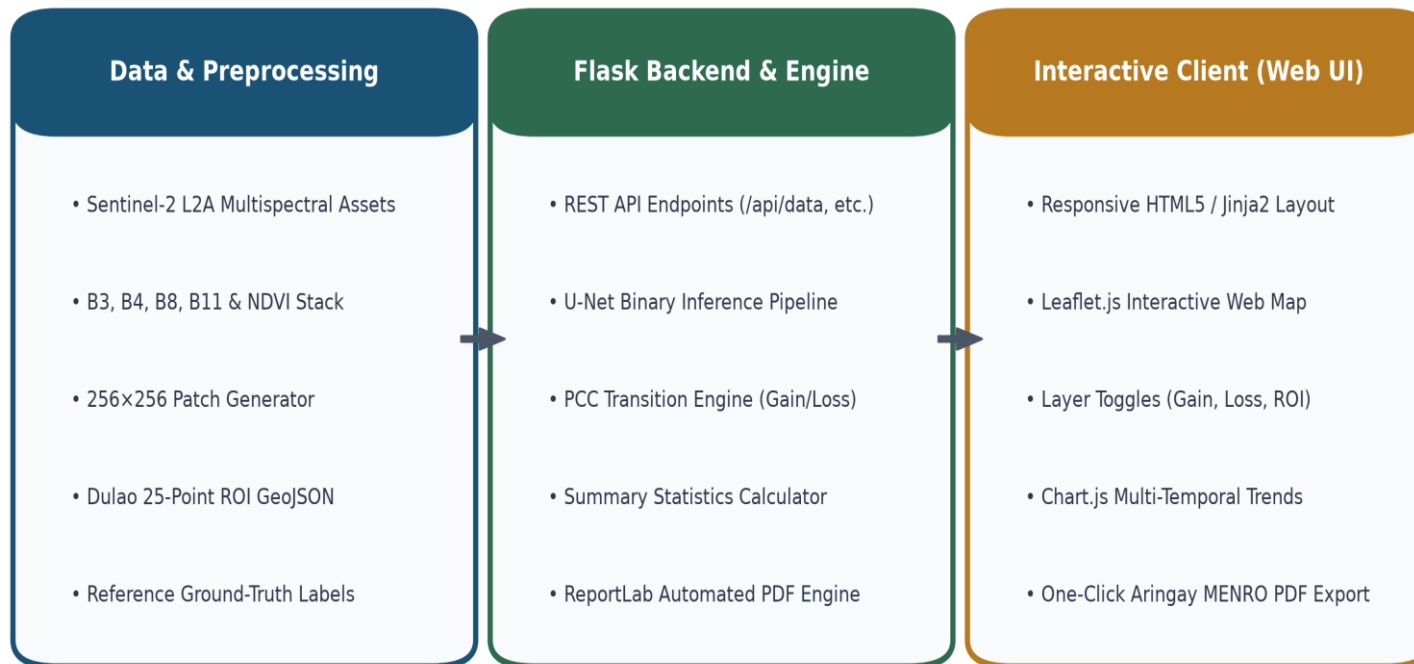
Mangrove in both T_1 & T_2 . Represents persistent core mangrove stands.

STABLE NON-MANGROVE (Value = 0):

Non-mangrove in both T_1 & T_2 . Open water, mudflats, sandbars, and upland soil.

Software Engineering: Three-Tier Prototype Architecture

Three-Tier Prototype Software Architecture



Clean Decoupled Architecture: Flask Backend communicates via REST JSON with Leaflet/Chart.js Frontend

SYSTEM HIGHLIGHTS

Decoupled Design:

Flask backend cleanly separated from Leaflet frontend via JSON REST endpoints.

REST API Layer:

/api/study-area, /api/statistics, and /api/change-detection endpoints.

Automated Reporting:

Headless ReportLab engine generates standardized A4 PDF reports on demand.

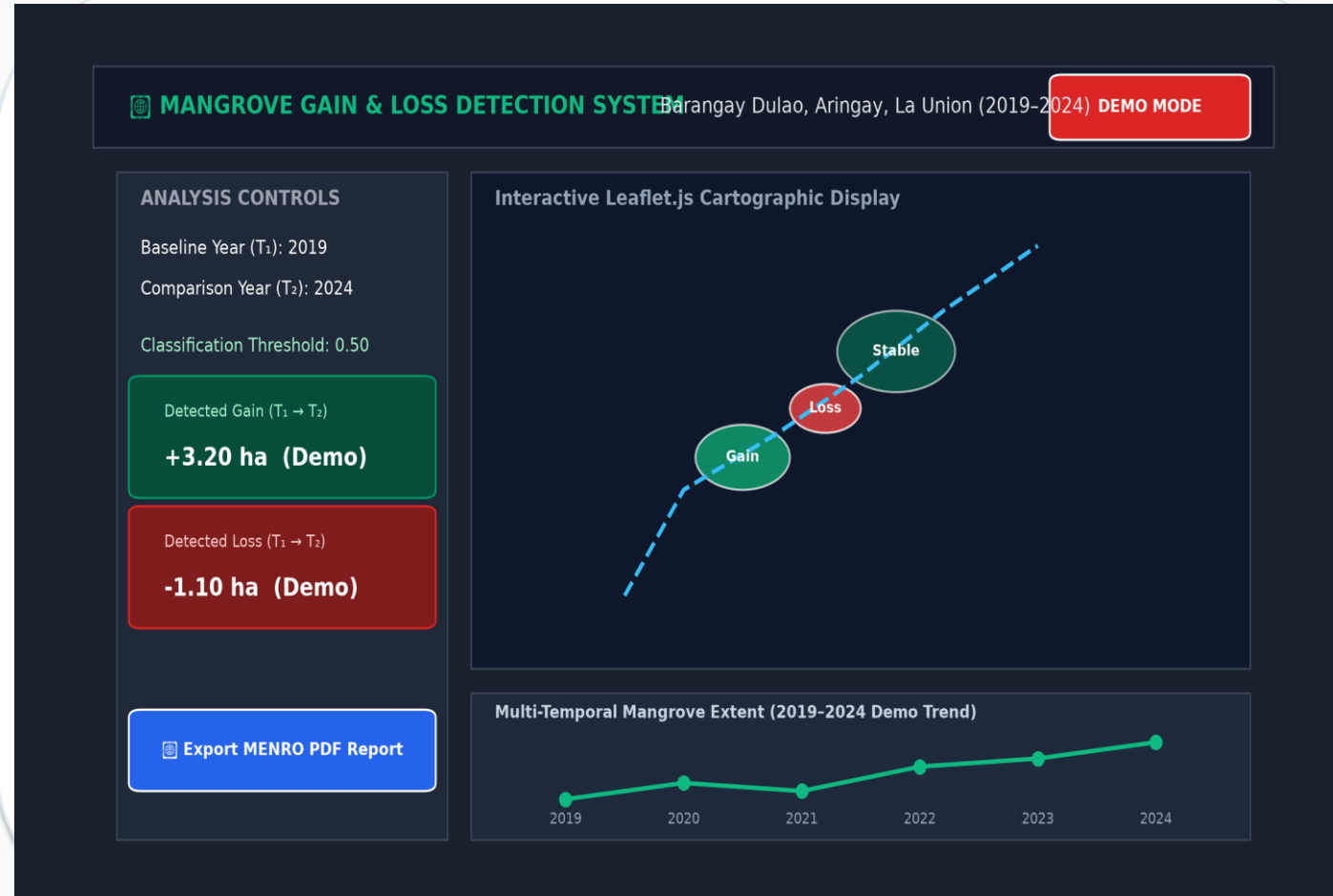
Interactive GIS:

Leaflet.js client renders multi-layer vector overlays with real-time opacity controls.

Production Readiness:

Config-driven architecture; switching DEMO_MODE to False immediately hooks real inference.

Prototype Implementation & User Interface Design



CORE FUNCTIONAL MODULES

Interactive Map Viewport:

Leaflet.js map displaying OpenStreetMap basemap, Dulao ROI boundary, and classified canopy layers.

Pairwise Temporal Wizard:

Dropdown selectors for T₁ (Baseline) and T₂ (Target), supporting 15 annual change comparisons.

Dynamic Metric Cards:

Immediate display of mapped mangrove extent (ha), net gain, net loss, and percentage change.

Multi-Temporal Trend Chart:

Chart.js line graph illustrating historical canopy trajectory across 2019–2024.

One-Click Report Export:

Generates formal municipal conservation reports formatted for Aringay MENRO.

Model Evaluation Metrics & Software Verification Protocol

ACADEMIC EVALUATION METRICS

Standard Segmentation Metrics

Precision:

$TP / (TP + FP)$ — Minimizes false positive mangrove claims over fishpond algae.

Recall (Sensitivity):

$TP / (TP + FN)$ — Ensures actual fragmented mangrove stands are not missed.

F1-Score (Dice Coeff):

$2 \times (Prec \times Rec) / (Prec + Rec)$ — Harmonic balance of accuracy across classes.

Intersection over Union (IoU):

$TP / (TP + FP + FN)$ — Standard geospatial overlap benchmark.

Mean Average Precision (mAP):

Integrated Area Under the Precision-Recall Curve across threshold sweeps.

Threshold Sweep Protocol:

Evaluated across 19 cutoffs from 0.05 to 0.95 (step 0.05) to determine optimal operating threshold.

SOFTWARE QUALITY VERIFICATION

Automated Test Assertions

Mathematical Partition Invariance:

Total ROI Area = Gain + Loss + Stable Mangrove + Stable Non-mangrove (verified across all 15 pairs).

Spatial Boundary Clamping:

All GeoJSON polygons strictly bounded within Dulao's 316.74-hectare envelope.

API Contract Verification:

JSON schema validation for /api/statistics, /api/data, and error handlers (HTTP 400 for invalid year pairs).

PDF Document Integrity:

ReportLab binary streams verified for valid EOF markers and correct table formatting.

Demonstration Label Transparency:

Explicit 'DEMO DATA' watermark displayed across all mock endpoints.

Live Prototype Demonstration & Operational Workflow

STEP 1

Select Baseline & Target

Choose baseline year T_1 (e.g. 2019) and comparison year T_2 (e.g. 2024) from intuitive dropdowns.

STEP 2

Execute Change Detection

Click 'Analyze Change' to trigger backend PCC matrix comparison and hectare calculations.

STEP 3

Interactive Cartography

Inspect color-coded gain/loss polygons in Leaflet, toggle layer overlays, and examine trend charts.

STEP 4

Generate MENRO Report

Click 'Export PDF Report' to automatically produce a formal, printable A4 summary document.

NOTICE FOR LIVE DEMO: The running system demonstrates full end-to-end functionality (routing, map rendering, spatial analytics, report generation) using controlled demonstration values to illustrate operational readiness.

Current Implementation Status & Project Scope Boundaries

System Component / Subsystem	Current Status	Implementation Mechanism	Academic & Operational Remarks
Web Frontend & Cartography	IMPLEMENTED	HTML5, CSS3, Leaflet.js, Chart.js	Interactive map, layer toggles, and multi-temporal charts fully operational.
Flask Backend & REST API	IMPLEMENTED	Python Flask WSGI, REST JSON	Routing, JSON payloads, and error handling fully implemented and tested.
Automated Report Engine	IMPLEMENTED	ReportLab Headless PDF Generator	Generates formatted, printable A4 MENRO summary reports dynamically.
Sentinel-2 Preprocessing	IMPLEMENTED	Google Earth Engine & Python Rasterio	Band isolation (B3, B4, B8, B11), bilinear resampling, and NDVI calculation.
5-Channel U-Net Architecture	IMPLEMENTED	PyTorch / Keras Deep Learning Code	Architecture definition, skip connections, loss functions, and sweep pipelines ready.
Independent Ground-Truth Validation	PENDING	Aringay MENRO Field Experts	Undergoing independent review by MENRO personnel to guarantee label validity.
Final Model Weights & Metrics	PENDING	Google Colab Pro GPU Training	Dependent on finalized MENRO-validated reference masks prior to final fit.
Live In-App Satellite Inference	OUT OF SCOPE	Pre-computed / Batch Inference	The prototype is designed for decision support rather than operational live satellite ingestion.

Interdisciplinary Synthesis & Tripartite Contribution

REMOTE SENSING

Earth Observation Pipeline

- Multi-temporal Sentinel-2 MSI data extraction.
- 10m spatial resolution across 2019–2024.
- 4 multispectral bands (B3, B4, B8, B11) + NDVI.
- Rigorous dry-season cloud filtration (< 20%).

DEEP LEARNING

Spatial-Contextual AI

- 5-channel modified U-Net architecture.
- Contextual feature extraction via skip connections.
- Overcomes traditional pixel-level spectral confusion.
- Threshold sweep optimization (0.05–0.95).

SOFTWARE ENGINEERING

Operational Decision Support

- Production-ready Flask REST backend.
- Interactive Leaflet.js cartographic client.
- Post-Classification Comparison (PCC) engine.
- One-click A4 PDF reporting for Aringay MENRO.

SYNTHESIS: Remote Sensing + Deep Learning + Software Engineering = An Integrated, Sustainable Municipal Mangrove Monitoring System

Conclusion & Immediate Implementation Roadmap

SUMMARY OF ACHIEVEMENTS

Current Project Milestones

End-to-End Pipeline Established:

Successfully demonstrated data ingestion from Sentinel-2, 5D tensor generation, and U-Net architecture definition.

Rigorous Software Prototype:

Flask WSGI application, REST API endpoints, Leaflet map, and ReportLab PDF generator 100% operational.

Mathematical Consistency:

PCC algorithm verified to preserve total area invariance (316.74 ha) across all 15 pairwise comparisons.

LGU-Centric Design:

Delivers an accessible, zero-cost decision support tool directly aligned with Aringay MENRO's operational workflow.

CONCRETE ROADMAP (NEXT STEPS)

Pathway to Final Deployment

1. Complete MENRO Validation:

Finalize formal review and independent ground-truth sign-off with Aringay MENRO forestry officers.

2. Execute GPU Training Run:

Train final 5-channel U-Net on Google Colab Pro GPU using validated labels.

3. Sweep Empirical Thresholds:

Execute 0.05–0.95 threshold evaluation to determine optimal operating F1-score and IoU.

4. Seamless Weight Integration:

Set DEMO_MODE = False in config to immediately hook empirical model inference into the prototype.

5. Municipal Handover & Training:

Conduct operational orientation with Aringay MENRO staff for regular annual monitoring.

Thank You Very Much.

Open for Questions, Defense Critiques, and Discussion

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DEFENSE QUICK REFERENCE GUIDE

- ✓ Study Area: Barangay Dulao, Aringay, La Union (316.74 ha)
- ✓ Temporal Baseline: 2019–2024 (Dry Season, Feb 1 – Apr 30)
- ✓ Sensors & Bands: Sentinel-2 MSI (B3, B4, B8, B11) + NDVI
- ✓ Model Architecture: 5-Channel U-Net (256×256×5 Tensor)
- ✓ Data Protocol: 80% Training / 20% Validation Partition
- ✓ Change Logic: Post-Classification Comparison (PCC)
- ✓ Current Prototype: Operational Flask/Leaflet with Demo Benchmarks